STAT 581: Admissibility and Stein's Phenomenon

Handwritten Notes 2

Liang-Cheng Chen

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Editorial conventions and sources

This English transcription follows `notes/stat581-note2.pdf`, pp. 1–8. Source p. 9 is blank. The reference is Yen-Chi Chen, *Lecture 1: Elementary decision theory*, Autumn 2026, `Lec1_decision.pdf`, Sections 1.5.1–1.5.2. Red boxes distinguish corrections from the handwritten argument; blue boxes give clarifications or additional derivations. There are no explicit handwritten `#TODO` markers in this file; the continuity question on p. 2 is addressed explicitly below. All formulas are displayed in mathematical environments.

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1 The one-dimensional normal sample mean is admissible

1.1 Statement and background

Source: p. 1; lecture Theorem 1.23, pp. 1-14–1-15.

Under known positive sampling variance, an unrestricted real mean, and squared-error loss, the sample mean is admissible:

$$X_1, \dots, X_n \overset{\text{iid}}{\sim} N(\theta, \sigma^2), \quad \theta \in \mathbb{R}, \quad L(\theta, a) = (\theta - a)^2, \quad \delta_0(X) = \bar{X}_n. \tag{1}$$

Its risk is constant and it is also minimax:

$$r(\theta, \delta_0) = \frac{\sigma^2}{n}. \tag{2}$$

Notes 1 proves minimaxity using a sequence of proper normal priors. Admissibility requires more: a limit of admissible Bayes rules need not itself be admissible.

Correction: The background statements need their model conditions

The handwritten claim "not Bayes unless the parameter has compact support" is not a general criterion. In this unrestricted normal location problem, the sample mean is generalized Bayes for an improper flat prior; the posterior mean for a proper finite-variance normal prior is a different, shrinking rule. Compact support is neither the missing assumption nor a general guarantee that the sample mean is Bayes.
The half-line example in Notes 1 shows that the sample mean can become inadmissible when the parameter space changes and real-valued actions are allowed. Thus admissibility must always specify the model, loss, and rule class. Here the parameter space is the entire real line.

1.2 A hypothetical improvement persists on an interval

Source: pp. 1–2.

Rescale the data and parameter so that the sampling variance is one. If a competing rule dominates the sample mean, then

$$r(\theta, \delta') \le \frac{1}{n} \quad \text{for all } \theta, \qquad r(\theta_*, \delta') < \frac{1}{n} \quad \text{for some } \theta_*. \tag{3}$$

Continuity of risk gives an interval of positive length and a fixed positive improvement on that interval:

$$\text{there exist } \varepsilon > 0,\ a < b \text{ such that } r(\theta, \delta') \le \frac{1}{n} - \varepsilon \quad \text{for all } \theta \in [a, b]. \tag{4}$$

Correction: Contradiction assumption and quantifier (source p. 2)

The source starts "assume the sample mean is admissible" but immediately posits a dominating rule. The contradiction assumption must be *inadmissibility*. Also, strict improvement at one point yields *some* positive improvement on a nearby interval, not improvement by every positive number.

Supplement: Answer to the margin question about smoothness

Smoothness of squared loss alone does not justify continuity of an expectation. In this normal model, domination gives finite risk at every parameter and supplies the needed local integrability.

First average the hypothetical rule conditional on the sample mean. Normal sufficiency makes this conditional averaging independent of the parameter, and Jensen's inequality cannot increase risk. A dominating rule can therefore be taken as a measurable function of

$$Y = \bar{X}_n \sim N(\theta, 1/n). \tag{5}$$

On any compact parameter interval, choose two parameter values strictly outside it. The normal densities for parameters in the interval are bounded by a constant times the sum of the two exterior densities. This follows from the exponential-family expression

$$p_\theta(y) = \sqrt{\frac{n}{2\pi}} \exp\left(-\frac{ny^2}{2} + n\theta y - \frac{n\theta^2}{2}\right). \tag{6}$$

Finite risk at the exterior parameters implies integrability of the squared estimator there. Consequently a constant times one plus the squared estimator, multiplied by those two densities, dominates the risk integrands throughout the compact interval. Dominated convergence makes the risk continuous. The strict gap in (3) then implies (4). This spells out the continuity step used in the lecture.

1.3 Normal-prior Bayes risk

Source: p. 2; lecture normal–normal formula and p. 1-15.

Take a proper normal prior with fixed mean and positive variance:

$$\theta \sim N(\mu, \rho^2). \tag{7}$$

Before specializing to unit sampling variance, the posterior mean and variance are

$$\delta_\rho(X) = \frac{\rho^{-2}}{\rho^{-2} + n\sigma^{-2}}\mu + \frac{n\sigma^{-2}}{\rho^{-2} + n\sigma^{-2}}\bar{X}_n, \tag{8}$$

$$v_\rho = \frac{1}{\rho^{-2} + n\sigma^{-2}}. \tag{9}$$

The Bayes risk of the posterior mean equals the expected posterior variance. Since that variance is constant in the observations,

$$b_\rho := r_{\pi_\rho}(\delta_\rho) = \mathbb{E}\big[\mathrm{Var}(\theta \mid X)\big] = \frac{1}{\rho^{-2} + n\sigma^{-2}}, \tag{10}$$

$$\sigma^2 = 1 \quad \Longrightarrow \quad b_\rho = \frac{1}{\rho^{-2} + n} = \frac{\rho^2}{1 + n\rho^2}. \tag{11}$$

The Bayes risk of the competitor and Bayes optimality give

$$c_\rho := r_{\pi_\rho}(\delta') = \int_{\mathbb{R}} r(\theta, \delta')\pi_\rho(\theta)\, d\theta, \qquad b_\rho \le c_\rho. \tag{12}$$

Correction: The Bayes-risk numerator (source pp. 2–4)

The handwritten Bayes-risk formula incorrectly includes prior precision in the numerator. The numerator must be one. Prior precision belongs in the weight of the prior mean, not in posterior variance. In particular, the correct Bayes risk tends to the reciprocal sample size as the prior variance grows. Every occurrence in the ratio below uses this corrected expression. The lecture's Example 1.17 supplies the correct formula.

1.4 The ratio contradiction

Source: pp. 3–4; lecture p. 1-15.

Both risks can be compared with the constant risk of the sample mean. Define a nonnegative improvement function and compute the Bayes gap:

$$h(\theta) = \frac{1}{n} - r(\theta, \delta') \ge 0, \tag{13}$$

$$\frac{1}{n} - b_\rho = \frac{1}{n} - \frac{\rho^2}{1 + n\rho^2} = \frac{1}{n(1 + n\rho^2)}. \tag{14}$$

Therefore

$$\frac{1/n - c_\rho}{1/n - b_\rho} = n(1 + n\rho^2)\left(\frac{1}{n} - c_\rho\right) \tag{15}$$

$$= \frac{n(1 + n\rho^2)}{\sqrt{2\pi}\rho} \int_{\mathbb{R}} h(\theta) \exp\left[-\frac{(\theta - \mu)^2}{2\rho^2}\right] d\theta \tag{16}$$

$$\ge \frac{n(1 + n\rho^2)\varepsilon}{\sqrt{2\pi}\rho} \int_a^b \exp\left[-\frac{(\theta - \mu)^2}{2\rho^2}\right] d\theta. \tag{17}$$

Because the interval is fixed and bounded, for all sufficiently large prior standard deviations the exponential factor is at least one half throughout the interval. Thus

$$\frac{1/n - c_\rho}{1/n - b_\rho} \ge \frac{n(1 + n\rho^2)\varepsilon(b - a)}{2\sqrt{2\pi}\rho} \longrightarrow \infty \qquad (\rho \to \infty). \tag{18}$$

In particular, eventually the ratio exceeds one. Since its denominator is positive, this implies

$$\frac{1}{n} - c_\rho > \frac{1}{n} - b_\rho \quad \Longrightarrow \quad c_\rho < b_\rho. \tag{19}$$

This contradicts (12). Hence no dominating rule exists, and the one-dimensional sample mean is admissible.

Supplement: Why a diffuse prior works in this contradiction

The prior mass of a fixed interval decreases on the order of the inverse prior standard deviation. A fixed improvement there therefore produces a Bayes-risk improvement on that order. In contrast, the gap between the sample mean's risk and the optimal normal-prior Bayes risk decreases on the order of inverse prior variance. The former eventually exceeds the latter, which is incompatible with Bayes optimality.
The proof uses a sequence of proper priors; it does not assert that an improper prior defines an ordinary Bayes risk.

1.5 What changes with dimension?

Source: p. 5; lecture pp. 1-15–1-16.

For the unrestricted normal mean with identity covariance and total squared-error loss, the usual estimator is admissible in dimensions one and two, but inadmissible in every dimension at least three. The two-dimensional assertion is quoted from the lecture; the preceding one-dimensional interval proof does not itself prove it.

2 Stein's phenomenon and the James–Stein estimator

2.1 Model, loss, and the proposed improvement

Source: p. 6; lecture Theorem 1.24 and Remark 1.25, p. 1-16.

Use a single multivariate observation to simplify notation:

$$X \sim N_d(\theta, I_d), \quad \theta \in \mathbb{R}^d, \quad d \geq 3, \qquad L(\theta, a) = \|a - \theta\|_2^2 = \sum_{j=1}^{d} (a_j - \theta_j)^2. \tag{20}$$

The conventional rule and its risk are

$$\delta_0(X) = X, \qquad r(\theta, \delta_0) = d. \tag{21}$$

The James–Stein rule is

$$\delta_{\mathrm{JS}}(x) = \begin{cases} \left(1 - \dfrac{d-2}{\|x\|_2^2}\right) x, & x \neq 0, \\ 0, & x = 0. \end{cases} \tag{22}$$

It uses all coordinates through their squared norm. The value at the origin makes it a fully specified rule; that event has probability zero under every parameter. We will show strict improvement in risk at every parameter value.

2.2 Risk decomposition

Source: p. 7; lecture equation (1.6), p. 1-17.

Expand squared error, keeping the cross-term sign explicit:

$$r(\theta, \delta_{\mathrm{JS}}) = \mathbb{E}_\theta \left\| \theta - X + \frac{d-2}{\|X\|_2^2} X \right\|_2^2 \tag{23}$$

$$= \mathbb{E}_\theta \|\theta - X\|_2^2 + 2(d-2)\mathbb{E}_\theta \left[\frac{(\theta - X)^\mathsf{T} X}{\|X\|_2^2} \right] + (d-2)^2 \mathbb{E}_\theta \left[\frac{1}{\|X\|_2^2} \right] \tag{24}$$

$$= r(\theta, \delta_0) + C + D, \tag{25}$$

where

$$C = -2(d-2)\mathbb{E}_\theta \left[\frac{(X - \theta)^\mathsf{T} X}{\|X\|_2^2} \right], \tag{26}$$

$$D = (d-2)^2 \mathbb{E}_\theta \left[\frac{1}{\|X\|_2^2} \right]. \tag{27}$$

The last term is positive; the cross term will be negative with twice its magnitude.

2.3 Stein's identity

Source: p. 7; lecture Lemma 1.26, p. 1-17.

For a normal vector and a sufficiently regular vector-valued function, Stein's identity is

$$Y \sim N_d(\mu, \sigma^2 I_d) \quad \Longrightarrow \quad \mathbb{E}[(Y - \mu)^\mathsf{T} g(Y)] = \sigma^2 \mathbb{E}[\nabla \cdot g(Y)]. \tag{28}$$

The divergence is

$$\nabla \cdot g(y) = \sum_{j=1}^{d} \frac{\partial g_j(y)}{\partial y_j}. \tag{29}$$

For smooth functions with sufficient integrability and vanishing boundary terms, this follows by integration by parts in each coordinate:

$$\frac{\partial p(y)}{\partial y_j} = -\frac{y_j - \mu_j}{\sigma^2} p(y), \qquad \int (y_j - \mu_j) g_j(y) p(y)\, dy = \sigma^2 \int \frac{\partial g_j(y)}{\partial y_j} p(y)\, dy. \tag{30}$$

Correction: Regularity is part of Stein's lemma

The handwritten derivative expectation alone is not a full set of hypotheses. A usable statement must also ensure appropriate absolute or weak differentiability, integrability of the terms, and valid integration by parts. The function used next is singular at the origin, so the application is justified by a smooth approximation below.

2.4 Compute the divergence and finish the proof

Source: p. 8; lecture pp. 1-17–1-18.

Away from the origin, use

$$g(x) = \frac{x}{\|x\|_2^2}, \qquad \frac{\partial g_j(x)}{\partial x_j} = \frac{1}{\|x\|_2^2} - \frac{2x_j^2}{\|x\|_2^4}. \tag{31}$$

Summing over coordinates gives

$$\nabla \cdot g(x) = \frac{d}{\|x\|_2^2} - \frac{2\sum_{j=1}^d x_j^2}{\|x\|_2^4} \tag{32}$$

$$= \frac{d-2}{\|x\|_2^2}. \tag{33}$$

Stein's identity at unit sampling variance therefore yields

$$\mathbb{E}_\theta\left[\frac{(X-\theta)^\mathsf{T} X}{\|X\|_2^2}\right] = (d-2)\mathbb{E}_\theta\left[\frac{1}{\|X\|_2^2}\right]. \tag{34}$$

Substitute into the cross term and combine:

$$C = -2(d-2)^2\mathbb{E}_\theta\left[\frac{1}{\|X\|_2^2}\right], \tag{35}$$

$$r(\theta, \delta_{\mathrm{JS}}) = d - (d-2)^2\mathbb{E}_\theta\left[\frac{1}{\|X\|_2^2}\right] < d = r(\theta, \delta_0). \tag{36}$$

For dimensions at least three, the inverse-norm expectation is finite and strictly positive. Thus the usual estimator is inadmissible.

Supplement: Why the singularity is harmless in dimensions at least three

The normal density is bounded near the origin. In polar coordinates the local integrability condition for the inverse squared norm is

$$\int_{\|x\|\le 1} \|x\|^{-2}\, dx = |S^{d-1}| \int_0^1 r^{d-3}\, dr < \infty \quad \Longleftrightarrow \quad d > 2. \tag{37}$$

Outside the unit ball the inverse squared norm is bounded by one. To apply Stein's

identity rigorously, use the smooth functions

$$g_\eta(x) = \frac{x}{\|x\|_2^2 + \eta}, \qquad \eta > 0, \qquad \nabla \cdot g_\eta(x) = \frac{d}{\|x\|_2^2 + \eta} - \frac{2\|x\|_2^2}{(\|x\|_2^2 + \eta)^2}. \tag{38}$$

Their divergence is bounded in absolute value, off the origin, by the inverse squared norm times a fixed dimension-dependent constant. The other side of the identity is bounded by

$$|(X - \theta)^\top g_\eta(X)| \le \frac{\|X - \theta\|_2}{\|X\|_2}, \tag{39}$$

which is integrable by Cauchy–Schwarz and the finite inverse-norm expectation. Dominated convergence as the smoothing parameter decreases to zero proves the identity used above.

2.5 Interpretation and extension to a sample mean

Supplement to source pp. 5–8; lecture Remarks 1.25 and 1.27.

For independent multivariate normal observations with known spherical covariance, rescale the sample mean to apply the result:

$$X_i \overset{\text{iid}}{\sim} N_d(\theta, \sigma^2 I_d), \qquad s^2 = \frac{\sigma^2}{n}, \tag{40}$$

$$\delta_{\text{JS},n}(\bar{X}_n) = \left(1 - \frac{(d-2)s^2}{\|\bar{X}_n\|_2^2}\right) \bar{X}_n \quad \text{away from the origin,} \tag{41}$$

$$r(\theta, \delta_{\text{JS},n}) = ds^2 - (d-2)^2 s^4 \mathbb{E}_\theta[\|\bar{X}_n\|_2^{-2}] < ds^2. \tag{42}$$

Set the estimator to zero at the origin. This establishes inadmissibility of the multivariate sample mean under total squared error for dimensions at least three.

Pooling the coordinates through a common shrinkage factor can reduce total mean squared error even when the observed coordinates are independent. The assertion concerns total loss: it does not assert that each coordinate's risk improves at every parameter. The usual estimator remains admissible for each separate scalar normal mean problem.

Dominating a rule does not prove one's own admissibility

The lecture notes that the ordinary James–Stein estimator is itself inadmissible. A familiar further improvement is its positive-part version:

$$\delta_{\text{JS}+}(x) = \max\left\{0, 1 - \frac{d-2}{\|x\|_2^2}\right\} x \quad (x \neq 0), \qquad \delta_{\text{JS}+}(0) = 0. \tag{43}$$

This additional formula is an editorial supplement, not a formula written in the source. No admissibility claim for the positive-part rule is made here. The handwritten proof establishes the strict domination of the usual estimator by the ordinary James–Stein rule.